

Abdelrahman Khalifa

Systems Software Engineer — Rust · C · Linux · GSoC 2026 (Linux Foundation)

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SUMMARY

Systems software engineer working in Rust, C and modern C++ on Linux. Completed Google Summer of Code 2026 with The Linux Foundation (OpenPrinting), building the COSMIC desktop's printer stack and landing 7 PRs upstream in cups-rs and libcups. Comfortable with FFI, IPC, async code and maintainer code review.

OPEN SOURCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust; design discussions, code review and documentation with OpenPrinting and System76 mentors

| github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Designed a printer-management stack as 5 Rust crates: `printers-core` (types, grouping queues into physical devices), `-client` (async varlink client), `-server` (CUPS, IPP, DNS-SD and Printer Application backend), `-ui` (iced/libcosmic) and `-app`.
- Exposed the backend over varlink IPC from `cosmic-settings-daemon`, so the C FFI code never loads in the Settings GUI process; the standalone app falls back to running the server in-process.
- Built a Printers page and an Add Printer flow for COSMIC Settings: live updates, print queue control, supply levels, and legacy printers set up through Printer Applications (PAPPL) instead of PPD drivers.
- Tested on real HP printers and Printer Applications (Ghostscript, Gutenprint, HPLIP); CI checks formatting, Clippy and tests on several architectures and runs CUPS, Avahi and `ippeveprinter` to test discovery.

Upstream contributions — OpenPrinting cups-rs (Rust) and libcups (C) | Jan 2026 – Oct 2026

- `cups-rs`: ported to `libcups3` with CUPS 2 and CUPS 3 builds and CI; added IPP management operations, Printer Application support, connections by IP/port, `lpoptions` support and FFI bindings for the `libcups3` DNS-SD API ([PRs #18–#22](#), merged).
- `libcups`: fixed crashes in `cupsDNSSDDelete()` and a leaked client in the Avahi DNS-SD code ([#164](#), [#165](#), merged).
- COSMIC: Printers page in Settings ([cosmic-settings #2182](#)) and the varlink printers server ([cosmic-settings-daemon #188](#)), in review.
- Winter of Code 2026: implemented the `lp` and `lpstat` commands in Rust on `cups-rs` ([PR #11](#)).

Contributor — Boa JavaScript engine (Rust), Rust Clippy, Zephyr RTOS | Mar 2024 – Dec 2025

- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion; with tests ([PR #4553](#)).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` ([PR #16214](#)).
- Zephyr RTOS: implemented the `IP_MULTICAST_LOOP` socket option across 6 networking modules, merged upstream ([commit b117036](#)).

EXPERIENCE

Embedded Systems Engineer — EKSON | Aug 2024 – Present

- Built a modern C++/Qt control application for motion simulators and optimized its performance by ~40% through asynchronous processing and fewer threads; reliable UART control.
- Developed Zephyr RTOS firmware for an ESP32-S3 3-DoF simulator, migrating from FreeRTOS for more deterministic behavior and production readiness.

Embedded Linux Systems Intern — ODC | Dec 2023 – Jan 2024

- Built Linux systems with Yocto and Buildroot; worked with kernel modules, device drivers, device trees, file systems and hardware flashing.

TECHNICAL SKILLS

Languages: Rust, C, C++ (modern C++, Qt), Python, Bash, Java

Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers

Tooling: Cargo, Clippy, `rustfmt`, Git/GitHub, CI, CMake, Make, Yocto, Buildroot, QEMU, testing, debugging, performance optimization, code review, documentation

Desktop & embedded: iced, libcosmic, Qt; Zephyr, FreeRTOS, ARM Cortex-M, ESP32

ACHIEVEMENTS

- Codeforces Expert (handle `_Khalifa`) and ECPC 2023 finalist (Egyptian Collegiate Programming Contest).
- Deloitte Software Engineering Mentorship (React Native, 2024); IEEE student chapter Treasurer (\$10k+ budget).

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Graduation project: AUTOSAR-based V2X vehicle prototype.